Sample Questions

1. Write a 16-bit control word for the following micro-operations:
   a) R5 ←R4 – R2
   b) R1 ← R2 XOR R4
   c) R6 ←R1+R4
   d) R4 ←R4 AND R2
   e) R3 ←SHL R5

1 2 3	4 5 6	7 8 9	10 11 12	13	14 15 16
A	B	D	F	C_{in}	H

) Control word

Binary code	Function of selection variables					
	A	B	D	F with $C_{in} = 0$	F with $C_{in} = 1$	H
0 0 0	Input data	Input data	None	$A, C \leftarrow 0$	$A + 1$	No shift
0 0 1	$R1$	$R1$	$R1$	$A + B$	$A + B + 1$	Shift-right, $I_R = 0$
0 1 0	$R2$	$R2$	$R2$	$A - B - 1$	$A - B$	Shift-left, $I_L = 0$
0 1 1	$R3$	$R3$	$R3$	$A - 1$	$A, C \leftarrow 1$	0's to output bus
1 0 0	$R4$	$R4$	$R4$	$A \vee B$	—	—
1 0 1	$R5$	$R5$	$R5$	$A \oplus B$	—	Circulate-right with C
1 1 0	$R6$	$R6$	$R6$	$A \wedge B$	—	Circulate-left with C
1 1 1	$R7$	$R7$	$R7$	$\bar{A}$	—	—

2. Draw the diagram of a 4-bit status register that has four status bits - carry, sign, zero and overflow flags.
   Find the four status bits performing the following operations of the two numbers of 8-bits :
   Number A: 100
   Number B: 120

   a. A + B
   b. A – B
   c. A XOR B
   d. A OR B
   e. A AND B